

SYDNEY TECHNICAL HIGH SCHOOL

HSC ASSESSMENT TASK 1

MATHEMATICS

EXTENSION 1

FEBRUARY 2004

Time Allowed: 70 minutes

General Instructions:

- Start each question on a new page
- Board approved calculators may be used
- Marks indicated are approximate only
- All necessary working should be shown
- Marks may not be awarded for messy or poorly arranged work.

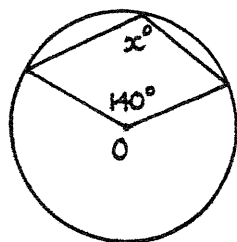
Name: _____

Class: _____

Q1	Q2	Q3	Q4	Q5	Q6	Total
/9	/9	/9	/7	/8	/9	/51

Question 1

(a)



O is the centre

Find x

Do not give reasons

1

(b)

Evaluate

$$\sum_{n=2}^6 2^n + 2n$$

2

(c)

i. Find the gradient of the tangent to the parabola $x^2 = 4ay$ at the point $P(2ap, ap^2)$

1

ii. Q is the point $(2aq, aq^2)$ and O is the origin. Show that if OQ is parallel to the tangent then $q = 2p$.

2

iii. If M is the midpoint of PQ, find the equation of the locus of M as P and Q vary along the parabola such that OQ remains parallel to the tangent at P.

3

Question 2

(a) Kerry deposits \$1500 into a superannuation fund on January 1st 2001. He makes further deposits of \$1500 on the first of each month up to and including December 1st 2010. The fund pays compound interest at a monthly rate of 0.75%. In each of the following questions give your answer to the nearest dollar.

i. How much is the first \$1500 deposit worth on December 31st 2010?

1

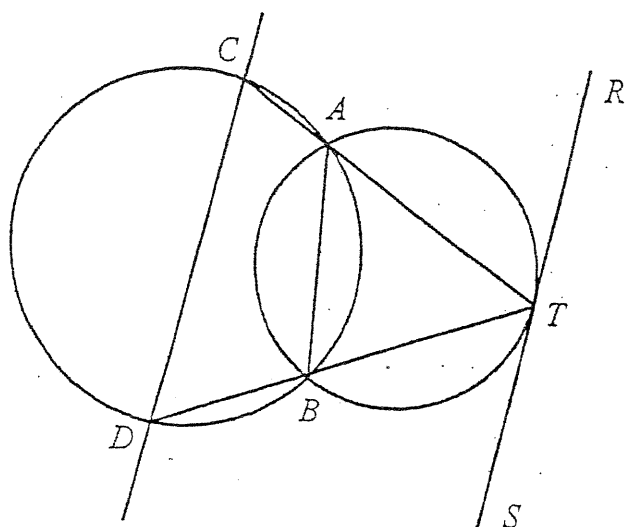
ii. Form a geometric series and hence determine the total amount in the fund on December 31st 2010.

2

iii. If each deposit was increased to \$1600, what difference does it make to the total amount in the fund on December 31st 2010?

2

(b)



In the diagram, two unequal circles intersect at A and B . The line RS is tangential to the smaller circle at T . The lines TA and TB meet the larger circle at C and D respectively.

- i. Copy the diagram
- ii. Explain why $\angle BAT = \angle BDC$ 1
- iii. Prove $RS \parallel CD$ 3

Question 3

- (a)
 - i. Expand $(n+1)^3$ 1
 - ii. Use the method of proof by induction to show that 4
 $1 + 7 + 19 + \dots + (3n^2 - 3n + 1) = n^3$ where n is a positive integer.
- (b) Three numbers whose product is 216 are in geometric progression. If 1, 4 and 8 are subtracted from them respectively the results are in arithmetic progression. Find the numbers. 4

Question 4

A caterer organises parties for groups of up to 200. She calculates the cost price of a party by allowing \$22 per head for the first 10 guests, \$21 per head for the next 10 guests, and so on, allowing one dollar less per head for each subsequent group of 10 guests or part thereof.

- i Show that the cost price, in dollars, for each guest in the n^{th} group of 10 guests, or part thereof, is given by 1

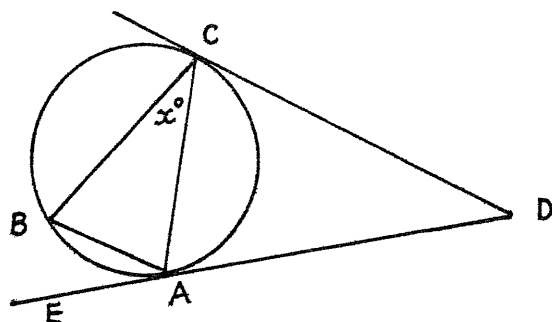
$$T_n = 23 - n$$

where T_n is the n^{th} term of an arithmetic series.

- ii. Find the increase in the cost price of a party if 4 more persons are added to a guest list of 85. 2
- iii. Determine the cost price of a party attended by 115 people. 2
- iv. If the caterer wishes to make a 25% profit on the cost price, calculate the average charge per head for a party of 115 guests. 2

Question 5

- (a) Use mathematical induction to show that if x is a positive integer then $(1+x)^n - 1$ is divisible by x for all positive integers $n \geq 1$ 4
- (b)



AD and CD are tangents to a circle. B is a point on the circle such that $\angle CBA$ and $\angle CDA$ are equal and are double $\angle BCA$.

Let $\angle BCA = x^\circ$

- i. Without adding any constructions find the value of x . Give reasons 2
- ii. Hence, prove that BC is a diameter of the circle. 2

Question 6

A fund is set up with a single investment of \$2000 to provide an annual prize of \$150. The fund accrues interest at 5% pa paid yearly. After the interest is added the prize money is withdrawn from the fund.

- i. Find the value of the fund immediately after the first prize has been awarded. 1
- ii. Show that the value of the fund after n years is given by 4
$$A_n = 3000 - 1000(1.05)^n$$
- iii. (a) In which year are there insufficient funds to award the full prize? 2
(b) For this final year, what is the maximum value of the prize that can be awarded. 2

Question 1

(a) $\underline{x = 110} \quad \text{--- ①}$

(b) $\sum_{n=2}^6 2^n + 2n$
 $= 2^2 + 2 \cdot 2 + 2^3 + 2 \cdot 3 + \dots + 2^6 + 2 \cdot 6$
 $= \underline{164} \quad \text{--- ①}$

(c) i. $y = \frac{x^2}{4a}$
 $\frac{dy}{dx} = \frac{2x}{4a}$
 $= \frac{x}{2a}$
 $m_{\text{tangent}} = \frac{2ap}{2a} = \underline{p} \quad \text{--- ①}$

ii. $m_{OQ} = \frac{aq^2 - 0}{2aq - 0}$
 $= \frac{q}{2} \quad \text{--- ①}$

OQ // tangent so
 $p = \frac{q}{2} \quad \text{--- ①}$
 $\therefore 2p = q$

iii. $M \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$
 $M \left(a(p+q), \frac{a(p^2 + q^2)}{2} \right)$

$x = a(p+q) \quad y = \frac{a(p^2 + q^2)}{2} \quad \text{--- ①}$

using $2p = q$

$x = a(p + 2p)$

$x = 3ap$

$p = \frac{x}{3a} \quad \text{--- ①}$

$y = \frac{a(p^2 + (2p)^2)}{2}$

$2y = 5ap^2$

$2y = 5a \times \left(\frac{x}{3a} \right)^2$

$2y = \frac{5ax^2}{9a^2}$

$\therefore \underline{18ay = 5x^2} \quad \text{--- ①}$

Question 2

(a) i. $A_1 = 1500 (1.0075)^{120}$
 $= \underline{\$ 3677} \quad \text{--- ①}$

ii. $A_2 = 1500 (1.0075)^{119}$
 $\vdots$
 $A_{120} = 1500 (1.0075)^1 \quad \text{--- ①}$

$A = 1500 (1.0075) + 1500 (1.0075) + \dots + 1500 (1.0075)$
 $= 1500 (1.0075 + \dots + 1.0075) \quad \text{--- ①}$

GP with $a = 1.0075$
 $r = 1.0075$
 $n = 120$

$= 1500 \times \frac{1.0075 (1.0075^{120} - 1)}{1.0075 - 1}$

$= \underline{\$ 292\,448} \quad \text{--- ①}$

iii. $A = 1600 \times \frac{1.0075 (1.0075^{120} - 1)}{1.0075 - 1}$

$= 311\,945 \quad \text{--- ①}$

$\therefore \text{difference is } \underline{\$ 19\,497} \quad \text{--- ①}$

iii. $\angle BAT = \angle BDC$ (above)
 $\angle BAT = \angle BTS$ (alternate segment theorem)

① $\therefore \angle BDC = \angle BTS$
① $\therefore RS \parallel CD$ (since alternate angles are equal)

$$(a) \quad i. \quad (n+1)^3 = (n+1)(n^2 + 2n + 1) \\ = \underline{\underline{n^3 + 3n^2 + 3n + 1}} \quad \text{--- (1)}$$

Step 1 : show true for $n=1$

$$RHS = 1^3$$

$\therefore$ true for $n=1$

$$S_k = k^3$$

ie show $S_{k+1} = (k+1)^3$ — (1)

$$\begin{aligned} S_{k+1} &= S_k + T_{k+1} \\ &= k^3 + [3(k+1)^2 - 3(k+1) + 1] \\ &= k^3 + 3k^2 + 6k + 3 - 3k - 3 + 1 \\ &= k^3 + 3k^2 + 3k + 1 \\ &= (k+1)^3 \end{aligned}$$

(b) let the numbers be

$$a, ar, ar^2 \leftarrow GP$$

$$a \times ar \times ar^2 = 216$$

$$(ar)^3 = 216 \quad \text{--- (1)}$$

$$ar = 6$$

$\therefore$ numbers are

a, b, br

$$a-1, 6-4, 6r-8 \leftarrow AP$$

$$a-1, 2, 6r-8$$

$$2 - (a-1) = 6r-8-2$$

$$3 - a = 6r - 10$$

$$a + 6r = 13$$

using $r = \frac{6}{a}$

$$a + 6.\underline{6} = 13$$

$$a^2 + 36^a = 13a$$

$$a^2 - 13a + 36 = 0$$

$$(a-9)(a-4) = 0$$

$$\begin{array}{lcl} a = 9 & \text{or} & a = 4 \\ r = \frac{2}{3} & & r = \frac{3}{2} \end{array}$$

$\therefore$ the numbers are 9, 6, 4
OR 4, 6, 9

Question 4

i. 22, 21, 20, ...

$$\begin{aligned} T_n &= a + (n-1)d \\ &= 22 + (n-1) \cdot -1 \\ &= 22 - n + 1 \end{aligned} \quad \text{--- (1)}$$

$$\therefore T_n = 23 - n$$

ii. for 85 guests $n = 9$

$$T_9 = 23 - 9 = 14 \text{ (per guest)}$$

$$\therefore \text{increase in cost} = 14 \times 4$$
$$= \underline{\$ 56} \quad \text{--- (1)}$$

iii. for the 111th-120th guest $n=12$

$$T_{12} = 23 - 12$$
$$= 11 \text{ (per guest) } \textcircled{1}$$

$$\begin{aligned}\text{Cost} &= 10 \times 22 + 10 \times 21 + \dots \\ &\quad + 10 \times 12 + 5 \times 11 \\ &= 10 [22 + 21 + \dots + 12] + 55 \\ &= 10 \times \frac{11}{2} [22 + 12] + 55 \\ &= \$ 1925 \quad \text{--- ①}\end{aligned}$$

$$\begin{array}{rcl} \text{iv.} & 125\% \times 1925 = & 2406.25 \\ & \underline{2406.25} & = \underline{\underline{\$ 20.93}} \\ & 115 & \end{array}$$

Question 5

(a) $(1+x)^n - 1$ divisible by x

Step 1 : show true for $n=1$

$$(1+x)^1 - 1 = x \quad \text{--- (1)}$$

which is divisible by x

Step 2 : assume true for $n=k$

$$(1+x)^k - 1 = Mx \quad (M \text{ is some integer})$$

①

Step 3: hence show true for $n = k+1$

ie show $(1+x)^{k+1} - 1$ is div. by

$$(1+x)^{k+1} - 1 \quad \text{--- (1)}$$

$$= (1+x)^k (1+x) - 1$$

$$= (Mx+1)(1+x) - 1$$

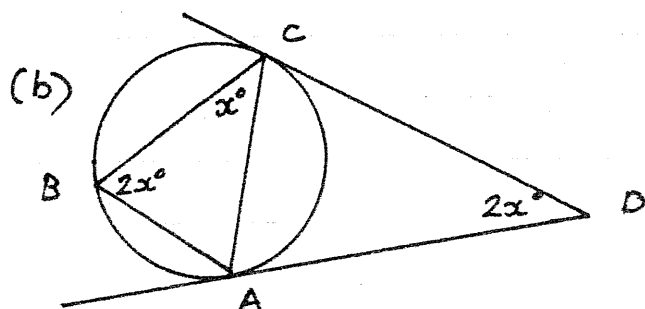
$$= Mx + Mx^2 + 1 + x - 1$$

$$= Mx + Mx^2 + x$$

$$= x(M + Mx + 1) \quad \text{--- ①}$$

which is div by x

Step 4 :



$$\begin{aligned} \angle CDA &= 2x \\ \angle CAD &= 2x \end{aligned} \quad \left(\begin{array}{l} \text{given} \\ \text{alternate segment} \\ \text{theorem} \end{array} \right)$$

i. $\angle CAD = 2x$ (alternate angles theorem)

$$\angle DCA = 2x \quad (\quad " \quad ")$$

$$6x = 180 \text{ (angle sum of } \triangle)$$

$$\therefore x = 30$$

ii. $\angle BCA = 30^\circ$
 $\angle CBA = 60^\circ$ —①
 $\therefore \angle CAB = 90^\circ$ (angle sum of Δ)
 $\therefore BC$ is diameter —①
 (angle in semi circle)
 is 90°

Question 6

i. $A_1 = 2000 + 0.05 \times 2000 - 150$
 $= 2000(1.05) - 150$
 $= \underline{\$1950}$ —①

ii. $A_2 = A_1 + 0.05 \times A_1 - 150$
 $= A_1(1.05) - 150$
 $= [2000(1.05) - 150]1.05 - 150$
 $= 2000(1.05)^2 - 150(1 + 1.05)$ —①

$A_n = 2000(1.05)^n - 150(1 + 1.05 + \dots + 1.05^{n-1})$ —①

GP with $a=1$
 $r=1.05$
 $n=n$

$= 2000(1.05)^n - 150 \times \frac{1(1.05^n - 1)}{1.05 - 1}$ —①

$= 2000(1.05)^n - \frac{150(1.05^n - 1)}{0.05}$

$= 2000(1.05)^n - 3000(1.05^n - 1)$ —①

$= 2000(1.05)^n - 3000(1.05)^n + 3000$

$\therefore A_n = 3000 - 1000(1.05)^n$

iii. (a) Prize can be awarded

$A_n \geq 0$

$3000 - 1000(1.05)^n \geq 0$ —①

$1000(1.05)^n \leq 3000$

$1.05^n \leq 3$

$n \leq 22.5$

$n = 22$

($A_{22} = 74.74$)

$\therefore$ insufficient funds in

22nd year —①

(b) $74.74 \times 1.05 = \underline{\$78.47}$

②

(-1 if interest not added)